Youwei Xiao

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Education

Bachelor of Science degree in Computer Science — September 2018 - July 2022
School of Electronics Engineering and Computer Science, Peking University — Beijing, China

Ph.D. Candidate in EDA — September 2022 - now
School of Integrated Circuits, Peking University — Beijing, China

- Research: compiler-driven hardware-software co-design for agile chips.

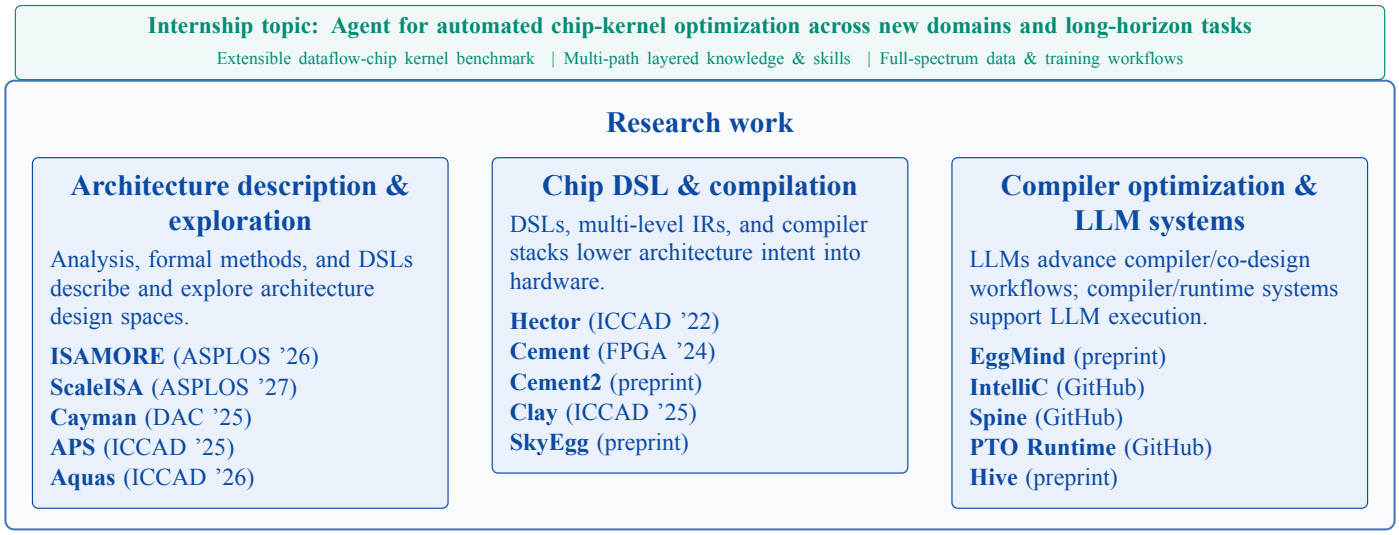
Work Experience

Large Language Model Algorithm Intern — June 2026 – Present
ByteDance Seed — Beijing, China

- Created an extensible benchmark of kernel programming and optimization tasks for dataflow-architecture chips.
- Developed a layered knowledge and skill system across multiple compilation paths and full-spectrum data collection and training workflows for continual model evolution.
- Pursuing the ultimate goal of developing a methodology and workflow that enable models and agents to rapidly acquire expert-level kernel optimization capabilities for new chip architectures.

Research

My research follows a compiler-driven approach to agile hardware-software co-design. Three coupled threads form a closed loop: architecture description and exploration discovers and formalizes reusable hardware capabilities; chip DSL and compilation express those capabilities and lower them into implementations; compiler optimization and LLM systems improve the resulting compiler, execution, and co-design workflows. My current work creates an extensible benchmark of kernel programming and optimization tasks for dataflow-architecture chips, develops a layered knowledge and skill system across multiple compilation paths, and builds full-spectrum data collection and training workflows for continual model evolution. The ultimate goal is a methodology and workflow that enable models and agents to rapidly acquire expert-level kernel optimization capabilities for new chip architectures.



Architecture description and exploration. I combine compiler analysis, formal methods, and architecture DSLs to represent and search design spaces while connecting customized hardware capabilities to programmable software. *ISAMORE* (DOI) discovers reusable custom instructions by anti-unifying equivalent program fragments; *ScaleISA*

(DOI, ASPLOS 2027) extends ISAMORE to MLIR and identifies scalable instructions with multi-level e-graph anti-unification; *Cayman* (DOI) identifies accelerator opportunities in complete applications while co-optimizing control flow and data access; *APS* (DOI, GitHub) and *Aquas* (ICCAD 2026) describe customized architectures through DSLs and expose them through complete hardware/software compiler stacks.

Chip DSL and compilation. I create DSLs and intermediate representations at multiple abstraction levels, then develop compilation and synthesis passes that optimize timing, microarchitecture, implementation selection, mapping, and scheduling. *Hector* (DOI, GitHub) provides multi-level MLIR foundations; *Cement* (DOI) and *Cement2* (preprint) introduce cycle-deterministic DSL and temporal-transaction abstractions; *Clay* (DOI) makes ASIP instructions microarchitecture-aware; *SkyEgg* (preprint) jointly optimizes implementation selection and scheduling with e-graphs.

Compiler optimization and LLM systems. I explore LLMs for compiler and hardware-software co-design workflows, together with compiler and system support for LLM workloads. *EggMind* (preprint) synthesizes equality-saturation strategies; *IntelliC* (GitHub) studies inspectable representations for human-model compilation; *Spine* (GitHub) organizes verification-bounded cross-layer co-synthesis; *PTO Runtime* (GitHub) supports dynamic kernel fusion and distributed execution of compiled task graphs on Ascend chips and LingQu SuperPods; *Hive* (preprint) provides multi-agent inference infrastructure.

Publications

- Yitian Sun, **Youwei Xiao**, Yuyang Zou, Chenyun Yin, and Yun Liang. 2027. Customizing Scalable Instructions via Multi-Level E-Graph Anti-Unification. In Proceedings of the 32nd ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS '27). doi:10.1145/3845814.3850048.
- **Youwei Xiao**, Chenyun Yin, Yitian Sun, and Yun Liang. 2026. Finding Reusable Instructions via E-Graph Anti-Unification. In Proceedings of the 31st ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS '26). doi:10.1145/3779212.3790162. **[Best Paper Award, 5/1048]**
- Yuyang Zou, **Youwei Xiao**, Yansong Xu, Chenyun Yin, Yuhao Luo, Yitian Sun, Ruifan Xu, Renze Chen, and Yun Liang. 2026. Aquas: Enhancing Domain Specialization through Holistic Hardware-Software Co-Optimization based on MLIR. In Proceedings of the 45th IEEE/ACM International Conference on Computer-Aided Design (ICCAD '26).
- **Youwei Xiao**, Yuyang Zou, Yansong Xu, Yuhao Luo, Yitian Sun, Chenyun Yin, Ruifan Xu, Renze Chen, and Yun Liang. 2025. APS: Open-Source Hardware-Software Co-Design Framework for Agile Processor Specialization. In Proceedings of the 44th IEEE/ACM International Conference on Computer-Aided Design (ICCAD '25). doi:10.1109/ICCAD66269.2025.11240817. [Invited Paper]
- Weijie Peng*, **Youwei Xiao***, Yuyang Zou, Zizhang Luo, and Yun Liang. 2025. Clay: High-level ASIP Framework for Flexible Microarchitecture-Aware Instruction Customization. In Proceedings of the 44th IEEE/ACM International Conference on Computer-Aided Design (ICCAD '25). doi:10.1109/ICCAD66269.2025.11240669. (*Equal contribution)
- **Youwei Xiao**, Fan Cui, Zizhang Luo, Weijie Peng, and Yun Liang. 2025. Cayman: Custom Accelerator Generation with Control Flow and Data Access Optimization. In Proceedings of the 62nd ACM/IEEE Design Automation Conference (DAC '25). doi:10.1109/DAC63849.2025.11132875.
- **Youwei Xiao**, Zizhang Luo, and Yun Liang. 2025. cmt2: Rule-Based Hardware Description in Rust with Temporal Semantics. In 5th Workshop on Languages, Tools, and Techniques for Accelerator Design (LATTE '25).
- Fan Cui, **Youwei Xiao**, Kexing Zhou, and Yun Liang. 2025. An Empirical Comparison of LLM-based Hardware Design and High-level Synthesis. In Proceedings of the 2025 ACM/SIGDA International Symposium on Field Programmable Gate Arrays (FPGA '25). doi:10.1145/3706628.3708861.
- Fan Cui, Chenyang Yin, Kexing Zhou, **Youwei Xiao**, Guangyu Sun, Qiang Xu, Qipeng Guo, Yun Liang, Xingcheng Zhang, Demin Song, and Dahua Lin. 2024. OriGen: Enhancing RTL Code Generation with Code-to-Code Augmentation and Self-Reflection. In Proceedings of the 43rd IEEE/ACM International Conference on Computer-Aided Design (ICCAD '24). doi:10.1145/3676536.3676830.
- **Youwei Xiao**, Zizhang Luo, Kexing Zhou, and Yun Liang. 2024. Cement: Streamlining FPGA Hardware Design with Cycle-Deterministic eHDL and Synthesis. In Proceedings of the 2024 ACM/SIGDA International Symposium on Field Programmable Gate Arrays (FPGA '24). doi:10.1145/3626202.3637561.
- Ruifan Xu, **Youwei Xiao**, Jin Luo, and Yun Liang. 2022. HECTOR: A Multi-Level Intermediate Representation for Hardware Synthesis Methodologies. In Proceedings of the 41st IEEE/ACM International Conference on Computer-Aided Design (ICCAD '22). doi:10.1145/3508352.3549370.

Preprints

- Chenyun Yin*, **Youwei Xiao***, Yuze Luo, Yuyang Zou, and Yun Liang. LLM-Guided Strategy Synthesis for Scalable Equality Saturation. arXiv:2604.17364. (*Equal contribution)
- Zizhang Luo, Yuhao Luo, **Youwei Xiao**, Yansong Xu, Runlin Guo, and Yun Liang. Hive: A Multi-Agent Infrastructure for Algorithm- and Task-Level Scaling. arXiv:2604.17353.
- **Youwei Xiao**, Yuyang Zou, and Yun Liang. SkyEgg: Joint Implementation Selection and Scheduling for Hardware Synthesis using E-graphs. arXiv:2511.15323.
- **Youwei Xiao**, Zizhang Luo, Weijie Peng, Yuyang Zou, and Yun Liang. Cement2: Temporal Hardware Transactions for High-Level and Efficient FPGA Programming. arXiv:2511.15073.

Academic Service

- **Program committees:** Student Technical Program Committee Member, MICRO 2026; Technical Program Committee Member, MLCAD 2026.
- **Peer reviewer:** IEEE TCAD (2024); ACM TECS (2024).

Honors and Awards

学术之芯 2026
Awarded to 8 students for academic excellence — School of IC, Peking University

博士研究生校长奖学金 2026-2027
Awarded to 10 students in School of IC, Peking University — Peking University

ASPLOS 2026 Best Paper Award March 2026
Finding Reusable Instructions via E-Graph Anti-Unification — 5/1048 submissions

Second Place in EDAthon 2020 August 2020
Programming Competition on Electronic Design Automation — IEEE CEDA Hong Kong Chapter

Shenzhen Stock Exchange Scholarship 2019-2020

Merit Student 2018-2019 & 2019-2020
Awarded to top 10% students for comprehensive excellence

Yang Fuqing & Wang Yangyuan Academician Scholarship 2018-2019
Awarded to only 5 students at EECS, Peking University, for academic excellence

Second Prize in NOI 2017 July 2017
National Olympiad in Informatics — China Computer Federation

Second Prize in APIO 2017 May 2017
Asia Pacific Informatics Olympiad (China District) — China Computer Federation

Tutorials

APS: An MLIR-Based Hardware-Software Co-design Framework January 2026
ASP-DAC 2026 — Hong Kong, China

Agile Hardware Specialization: A toolbox for Agile Chip Front-end Design May 2025
ISEDA 2025 — Hong Kong, China

Agile Hardware Specialization (AHS) March 2025
ASPLOS 2025 — Rotterdam, Netherlands

Agile Hardware Specialization: A toolbox for Agile Chip Front-end Design March 2025
DATE 2025 — Lyon, France

AHS: An EDA toolbox for Agile Chip Front-end Design January 2025
ASP-DAC 2025 — Tokyo, Japan